The Positivity Conjecture for $Q_{n,c}(q)$

Reference summary for research on cylindric partitions

February 10, 2026

Abstract

This document collects the statement, context, and motivation for the positivity conjecture on the polynomials $Q_{n,c}(q)$ arising from the two-variable generating function of rank-3 cylindric partitions. The conjecture first appears as Conjecture 4.2 of Corteel–Dousse–Uncu [4] and independently as Conjecture 2.7 of Warnaar [11]. We give the relevant definitions from the theory of cylindric partitions, state the conjecture precisely, and illustrate it with examples.

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1 Motivation: from Rogers–Ramanujan to A_2

1.1 The Rogers–Ramanujan identities

The Rogers–Ramanujan identities [10]

$$\sum_{n\geq 0}\frac{q^{n^2}}{(q;q)_n}=\prod_{n\geq 0}\frac{1}{(1-q^{5n+1})(1-q^{5n+4})}, \tag{1}$$

$$\sum_{n\geq 0}\frac{q^{n^2+n}}{(q;q)_n}=\prod_{n\geq 0}\frac{1}{(1-q^{5n+2})(1-q^{5n+3})}, \tag{2}$$

are among the most celebrated q-series identities. They have interpretations in the theory of partitions, the representation theory of affine Lie algebras, conformal field theory, and statistical mechanics.

1.2 The Andrews–Gordon identities

Gordon [9] generalised the combinatorial form of the Rogers–Ramanujan identities to arbitrary odd moduli $2k+1$. Andrews [1] found the analytic counterpart: for $r \geq 2$ and $1 \leq s \leq r$,

$$\sum_{n_1 \geq n_2 \geq \cdots \geq n_{r-1} \geq 0} \frac{q^{n_1^2+\cdots+n_{r-1}^2+n_s+\cdots+n_{r-1}}}{(q;q)_{n_1} \prod_{i=1}^{r-2} \begin{bmatrix} n_i \\ n_{i+1} \end{bmatrix}} = \frac{(q^{2r+1};q^{2r+1})_\infty}{(q;q)_\infty}\, \theta(q^s;q^{2r+1}). \tag{3}$$

The cases $r=2$, $s \in \{1,2\}$ recover (1)–(2). These identities correspond to principally specialised characters of the affine Lie algebra $\mathrm{A}_1^{(1)}$.

1.3 The A_2 problem

Finding higher-rank generalisations of the Andrews–Gordon identities for $\mathrm{A}_{r-1}^{(1)}$ with $r \geq 3$ has been a long-standing open problem. Andrews, Schilling and Warnaar [2] discovered several A_2 Rogers–Ramanujan identities (moduli 4 and 7) using an A_2 Bailey lemma, but the resulting q-series were expressed as characters multiplied by a factor of $(q;q)_\infty$ that obscured positivity.

The breakthrough insight, due to Foda–Welsh [7] and Corteel [5], was that **cylindric partitions** provide a natural combinatorial framework for these identities. The key observation is that the generating function for cylindric partitions of rank 3, when evaluated at $z=1$, yields the A_2 Andrews–Gordon characters (up to a universal factor).

This raises a fundamental question: can one find *manifestly positive* multisum expressions for all A_2 Andrews–Gordon identities? Corteel, Dousse and Uncu [4] made significant progress on this for level $d=5$, and the general positivity conjecture emerged from their work and that of Warnaar [11].

2 Cylindric partitions

2.1 Definition

Definition 1 (Cylindric partition, rank 3)**.** Let $c=(c_0,c_1,c_2)$ be a composition of $d := c_0+c_1+c_2$. A *cylindric plane partition of profile* c is a triple of partitions $\nu = (\nu^{(1)},\nu^{(2)},\nu^{(3)})$ satisfying the interlacing conditions:

$$\nu_j^{(1)} \geq \nu_{j+c_1}^{(2)} \qquad \text{for all } j \geq 1, \tag{4}$$

$$\nu_j^{(2)} \geq \nu_{j+c_2}^{(3)} \qquad \text{for all } j \geq 1, \tag{5}$$

$$\nu_j^{(3)} \geq \nu_{j+c_0}^{(1)} \qquad \text{for all } j \geq 1. \tag{6}$$

Here $\nu_j^{(i)} = 0$ whenever j exceeds the length of $\nu^{(i)}$. The *size* and *max* statistics are

$$|\nu| := |\nu^{(1)}| + |\nu^{(2)}| + |\nu^{(3)}|, \qquad \max(\nu) := \max\bigl(\nu_1^{(1)},\, \nu_1^{(2)},\, \nu_1^{(3)}\bigr).$$

Remark 2. This is a specialisation of the general definition of Gessel–Krattenthaler [8] to rank $r=3$. The profile c encodes the "shifts" in the interlacing conditions; equivalently, the cylindric partition lives on a cylinder of circumference 3 with profile determining the shape of the fundamental domain.

Example 3 (Profile $(2,1)$)**.** For $c=(2,1,0)$ (so $d=3$), a cylindric partition is a triple $(\nu^{(1)},\nu^{(2)},\nu^{(3)})$ where:

- $\nu_j^{(1)} \geq \nu_{j+1}^{(2)}$ (shift by $c_1 = 1$),
- $\nu_j^{(2)} \geq \nu_j^{(3)}$ (shift by $c_2 = 0$),
- $\nu_j^{(3)} \geq \nu_{j+2}^{(1)}$ (shift by $c_0 = 2$).

2.2 The generating functions

Definition 4 (Two-variable generating function)**.** The bivariate generating function for cylindric partitions with profile c is

$$\mathrm{GK}_c(z,q) := \sum_{\nu \in \mathrm{CPP}(c)} z^{\max(\nu)}\, q^{|\nu|}, \tag{7}$$

where the sum runs over all cylindric partitions of profile c.

Definition 5 (Size generating function)**.** The (univariate) size generating function is obtained by setting $z = 1$:

$$\mathrm{GK}_c(q) := \mathrm{GK}_c(1,q) = \sum_{\nu \in \mathrm{CPP}(c)} q^{|\nu|}.$$

2.3 Borodin's product formula

Theorem 6 (Borodin [3])**.** *The size generating function factors as an infinite product. For rank 3 with profile $c = (c_0, c_1, c_2)$ and $d = c_0 + c_1 + c_2$,*

$$\mathrm{GK}_c(q) = \frac{1}{(q;q)_\infty} \cdot \frac{1}{\displaystyle\prod_{\square \in \mu} (q^{h(\square)};q^d)_\infty \cdot \prod_{\square \in \mu^c} (q^{d-h(\square)};q^d)_\infty},$$

where $\mu = \mu(c)$ is the partition associated to the profile, $h(\square)$ is the hook length, and μ^c is the complement of μ in the $3 \times s(1,3)$ rectangle, with $s(i,j) = c_i + c_{i+1} + \cdots + c_{j-1}$.

2.4 Connection to A_2 characters

The crucial link to representation theory is the identity (Foda–Welsh [7]):

$$\mathrm{AG}_{\lambda,3}(q) = (q;q)_\infty \cdot \mathrm{GK}_c(q), \tag{8}$$

where $\mathrm{AG}_{\lambda,3}(q)$ is the A_2 Andrews–Gordon series associated to the $\mathrm{A}_2^{(1)}$ weight $\lambda = (d - \mu_1 + \mu_3)\Lambda_0 + (\mu_1 - \mu_2)\Lambda_1 + (\mu_2 - \mu_3)\Lambda_2$, and the profile entries are $c_0 = d - \mu_1 + \mu_3$, $c_i = \mu_i - \mu_{i+1}$ for $1 \leq i \leq 2$. This means that every identity about $\mathrm{GK}_c(q)$ translates to a statement about specialised affine Lie algebra characters.

3 The E–H factorisation and the polynomials $Q_{n,c}(q)$

3.1 Factoring out the universal series

The generating function $\mathrm{GK}_c(z,q)$ contains a "universal" factor independent of the profile. Defining the corrected generating function

$$G_c(z,q) := (zq;q)_\infty \cdot \mathrm{GK}_c(z,q), \tag{9}$$

the factor $H(z) = 1/(zq;q)_\infty = \sum_{n\geq 0} q^n z^n/(q;q)_n$ (representing ordinary partitions with largest part $\leq n$, or equivalently with at most n parts; see Section **??** below) is stripped off. Note that $\mathrm{GK}_c(z,q) = H(z)\cdot G_c(z,q)$ is the E–H factorisation of the cylindric partition generating function.

3.2 Why z tracks the maximum entry

The variable z in $\mathrm{GK}_c(z,q) = \sum z^{\max(\mathfrak{c})}\, q^{|\mathfrak{c}|}$ tracks the maximum entry $\max(\mathfrak{c}) = \max_k \mu_1^k$. This is natural because the universal factor $H(z)$ admits two equivalent readings, related by the conjugation involution $\lambda \mapsto \lambda'$ on partitions:

$$H(z) = \frac{1}{(zq;q)_\infty} = \sum_{\lambda} z^{\lambda_1}\, q^{|\lambda|} = \sum_{\lambda} z^{\ell(\lambda)}\, q^{|\lambda|}, \tag{10}$$

where the first sum tracks the *largest part* λ_1 and the second tracks the *number of parts* $\ell(\lambda)$. These are the same generating function because conjugation preserves size and swaps $\lambda_1 \leftrightarrow \ell(\lambda)$.

- **Height reading** (z^{λ_1}): the z-degree measures the height of the tallest column. This is the statistic that appears in the definition of $\mathrm{GK}_c(z,q)$.
- **Depth reading** ($z^{\ell(\lambda)}$): the z-degree measures the number of parts, which in the growth diagram equals the number of peeling layers needed to decompose the CPP.

The E–H factorisation then expresses $z^{\max(\mathfrak{c})}$ as a convolution of the base partition's contribution (height or depth of γ) and the purely cylindric contribution from $G_c(z,q)$.

3.3 The corrected coefficients

Write the z-expansion of the corrected generating function as

$$G_c(z,q) = \sum_{n\geq 0} g_{n,c}(q)\, z^n, \tag{11}$$

where $g_{n,c}(q) = [z^n]\, G_c(z,q)$. By the Euler expansion $(zq;q)_\infty = \sum_{j\geq 0}(-1)^j q^{j(j+1)/2} z^j/(q;q)_j$, we have the alternating sum expression:

$$g_{n,c}(q) = \sum_{j=0}^{n} \frac{(-1)^j\, q^{j(j+1)/2}}{(q;q)_j} \cdot [z^{n-j}]\ \mathrm{GK}_c(z,q). \tag{12}$$

3.4 The polynomial $Q_{n,c}(q)$

Definition 7 ($Q_{n,c}(q)$)**.** For a nonneg integer n and profile $c = (c_0, c_1, c_2)$ with $d = c_0 + c_1 + c_2$ and $\ell = \gcd(d, 3)$, define

$$Q_{n,c}(q) := (q^\ell;q^\ell)_n \cdot [z^n]\Big((zq;q)_\infty \cdot \mathrm{GK}_c(z,q)\Big) = (q^\ell;q^\ell)_n \cdot g_{n,c}(q). \tag{13}$$

When $d \not\equiv 0 \pmod 3$, we have $\ell = 1$ and $Q_{n,c}(q) = (q;q)_n \cdot g_{n,c}(q)$.

Remark 8. The multiplication by $(q^\ell;q^\ell)_n$ clears the denominators appearing in (11), converting the formal power series $g_{n,c}(q)$ into a polynomial. That $Q_{n,c}(q)$ is indeed a polynomial (and not merely a formal series) was proved by Welsh [12].

3.5 The q-binomial decomposition

The definition yields a useful expansion in terms of q-binomial coefficients. Write $f_{m,c}(q) := [z^m]\ \mathrm{GK}_c(z,q)$ for the z-coefficients of the *original* (uncorrected) generating function, and define

$$h_{m,c}(q) := (q;q)_m \cdot f_{m,c}(q). \tag{14}$$

Note that $h_{m,c}(q) \neq Q_{m,c}(q)$: the former uses the coefficients $f_{m,c}$ of $\mathrm{GK}_c(z,q)$, while the latter uses the coefficients $g_{m,c}$ of the corrected $G_c(z,q) = (zq;q)_\infty \cdot \mathrm{GK}_c(z,q)$.

Then:

$$Q_{n,c}(q) = \sum_{j=0}^{n} (-1)^j\, q^{j(j+1)/2} \begin{bmatrix} n \\ j \end{bmatrix} \cdot h_{n-j,c}(q). \tag{15}$$

This follows from expanding $(zq;q)_\infty$ by the Euler identity $(zq;q)_\infty = \sum_j (-1)^j q^{j(j+1)/2} z^j/(q;q)_j$, extracting $[z^n]$, and multiplying through by $(q;q)_n$, using $(q;q)_n/(q;q)_j = \begin{bmatrix} n \\ j \end{bmatrix} \cdot (q;q)_{n-j}$.

Computational evidence shows that $h_{m,c}(q)$ is itself a polynomial with nonneg coefficients for all tested profiles. This is a *stronger* observation than positivity of $Q_{n,c}(q)$, since $Q_{n,c}(q)$ is built from the $h_{m,c}(q)$ via an alternating-sign sum.

4 The conjecture

We now state the positivity conjecture. The formulation below combines the statements of Corteel–Dousse–Uncu [4, Conjecture 4.2] and Warnaar [11, Conjecture 2.7].

Conjecture 9 (Corteel–Dousse–Uncu; Warnaar)**.** *Let $c = (c_0, c_1, c_2)$ and $d := c_0 + c_1 + c_2$ with $d \not\equiv 0 \pmod 3$. Then:*

*(i) $Q_{n,c}(q)$ is a polynomial in q with **nonneg coefficients**.*

(ii) The evaluation at $q = 1$ is

$$Q_{n,c}(1) = \left(\frac{(d+1)(d+2)}{6} - 1 \right)^n. \tag{16}$$

Remark 10 (Status)**.** The polynomiality of $Q_{n,c}(q)$ and the $q = 1$ evaluation (15) have been **proved** by Welsh [12]. The positivity (i) **remains open** in general.

Remark 11 (The $d \equiv 0 \pmod 3$ case)**.** When $3 \mid d$ the appropriate normalisation uses $\ell = 3$, and the polynomial $P_{n,c}(q) := (q^3;q^3)_n \cdot g_{n,c}(q)$ is conjectured to have nonneg coefficients as well (Corteel–Dousse–Uncu [4, Conjecture 4.2]).

Remark 12 (Equivalent formulation via $g_{n,c}$)**.** Writing $g_{n,c}(q) = Q_{n,c}(q)/(q;q)_n$ (when $\ell = 1$), the conjecture says that the generating function of cylindric partitions with max entry exactly n, after correction by $(zq;q)_\infty$, has the form "nonneg polynomial divided by $(q)_n$". In other words, the alternating-sign cancellation in (11) resolves perfectly into positivity after clearing denominators.

5 Examples

5.1 Level $d = 2$: the Rogers–Ramanujan identities

For $d = 2$ there are two profiles (up to the shift symmetry $\mathrm{GK}_{(c_0,c_1,c_2)}(z,q) = \mathrm{GK}_{(c_1,c_2,c_0)}(z,q)$):

Profile $c = (2, 0, 0)$**.** The interlacing conditions force $\nu^{(2)} = \nu^{(3)} = \emptyset$ and $\nu^{(1)}$ is any partition with parts $\leq n$ (when $\max = n$). Here

$$Q_{n,(2,0,0)}(q) = q^{n^2},$$

a single monomial. Combining with the E–H factorisation and Borodin's formula reproduces the **first Rogers–Ramanujan identity** (1):

$$\mathrm{GK}_{(2,0,0)}(1, q) = \frac{1}{(q;q)_\infty} \cdot \frac{1}{(q^2;q^5)_\infty \cdot (q^3;q^5)_\infty}, \qquad \mathrm{AG}_{\Lambda_0+\Lambda_1;3}(q) = \sum_{n\geq 0} \frac{q^{n^2}}{(q;q)_n}.$$

Profile $c = (1, 1, 0)$**.** Here

$$Q_{n,(1,1,0)}(q) = q^{n(n+1)},$$

again a single monomial, and the identity becomes the **second Rogers–Ramanujan identity** (2):

$$\mathrm{AG}_{2\Lambda_0;3}(q) = \sum_{n\geq 0} \frac{q^{n^2+n}}{(q;q)_n}.$$

In both cases the conjecture holds trivially since a monomial has nonneg coefficients.

5.2 Level $d = 4$: the Andrews–Schilling–Warnaar identities

For $d = 4$ we have $\frac{1}{6}(5)(6) - 1 = 4$, so $Q_{n,c}(1) = 4^n$.

Profile $c = (2, 2, 0)$**.** The first few polynomials are:

$$\begin{aligned}
Q_{0,(2,2,0)}(q) &= 1,\\
Q_{1,(2,2,0)}(q) &= q^2 + q^3 + q^4 + q^5,\\
Q_{2,(2,2,0)}(q) &= q^4 + q^5 + 3q^6 + 3q^7 + 5q^8 + 4q^9 + 5q^{10} + 3q^{11} + 3q^{12} + q^{13} + q^{14}.
\end{aligned}$$

These are manifestly positive and palindromic, with $Q_{1,(2,2,0)}(1) = 4$.

Profile $c = (2, 1, 1)$**.** The generating function $\mathrm{GK}_{(2,2,1)}(z, q)$ was computed by Corteel–Dousse–Uncu [4, Theorem 3.3]. Their result yields:

$$Q_{n,(2,2,1)}(q) = \sum_{\substack{n_2,n_3,n_4\geq 0\\ m_1,\ldots,m_{k-1}\geq 0}} q^{n^2+n_2^2+n_3^2+n_4^2-nn_2+n_2n_4} \begin{bmatrix} n \\ n_2 \end{bmatrix} \begin{bmatrix} n \\ n_4 \end{bmatrix} \begin{bmatrix} n_2 \\ n_3 \end{bmatrix},$$

which gives $Q_{n,(2,2,1)}(1) = 6^n$. This is a manifestly positive quadruple sum.

5.3 Level $d = 5$: the Corteel–Dousse–Uncu identities

For $d = 5$ we have $\frac{1}{6}(6)(7) - 1 = 6$, so $Q_{n,c}(1) = 6^n$. This is the main case treated in [4], where manifestly positive multisum expressions for $\mathrm{GK}_c(z, q)$ were found for all five profiles (up to symmetry).

5.4 Level $d = 7$

For $d = 7$ we have $\frac{1}{6}(8)(9) - 1 = 11$, so $Q_{n,c}(1) = 11^n$. There are four distinct profiles up to cyclic shift: $(3, 2, 2)$, $(2, 3, 2)$, $(4, 2, 1)$, and $(5, 1, 1)$. Computational verification confirms positivity of $Q_{n,c}(q)$ for all profiles with $n \leq 3$.

6 Why cylindric partitions?

6.1 The product side

Borodin's formula (Theorem 6) gives a closed-form *infinite product* for $\mathrm{GK}_c(1, q)$. Via equation (8), this product is (up to the factor $1/(q;q)_\infty$) an $\mathrm{A}_2^{(1)}$ character. So each profile c corresponds to a specific affine Lie algebra module, and the generating function of cylindric partitions is its character.

6.2 The sum side

The conjecture posits that the *sum side*—the multisum expression for $Q_{n,c}(q)$—has nonneg coefficients. Finding these multisum expressions amounts to finding **manifestly positive** formulas for A_2 Andrews–Gordon identities.

The connection is:

$$\mathrm{AG}_{\lambda,3}(q) = (q;q)_\infty \cdot \mathrm{GK}_c(1, q) = \sum_{n \geq 0} \frac{Q_{n,c}(q)}{(q;q)_n}. \tag{17}$$

So positivity of $Q_{n,c}(q)$ would give a manifestly positive "sum = product" identity for every A_2 Andrews–Gordon character (at levels not divisible by 3).

6.3 The role of the max statistic

The key innovation is the *two-variable* generating function $\mathrm{GK}_c(z, q)$, which refines the character by the max statistic. Without the variable z, one only sees the total count $\mathrm{GK}_c(1, q)$ and the connection to $Q_{n,c}(q)$ is lost. The max statistic provides the z-grading that decomposes the character into the pieces $g_{n,c}(q) = [z^n]\, G_c(z, q)$, each of which carries the positivity.

This was first observed by Corteel–Welsh [6], who showed that the Corteel–Welsh q-difference equation (Theorem 2.3 of [4]) for $\mathrm{GK}_c(z, q)$ can be solved recursively to produce manifestly positive expressions when d is small.

7 The hierarchy of generating functions

The generating function $\mathrm{GK}_c(z, q)$ that appears in the conjecture is the simplest in a hierarchy of progressively more refined generating series for cylindric partitions. Each level specialises to the one below it, and the conjecture lives at the bottom of the chain. Understanding the richer levels may be the key to proving positivity.

7.1 Level 0: the univariate generating function

The coarsest invariant tracks only the total size. For a binary string π of length T encoding the profile, Borodin's identity [3] gives:

$$\sum_{\mathfrak{c}\in\mathrm{CPP}(\pi)} z^{|\mathfrak{c}|} = \prod_{n\geq 0}\left(\frac{1}{1-z^{(n+1)T}}\prod_{\substack{i<j\\ \pi_i>\pi_j}}\frac{1}{1-z^{j-i+nT}}\prod_{\substack{i>j\\ \pi_i>\pi_j}}\frac{1}{1-z^{j-i+(n+1)T}}\right). \tag{18}$$

7.2 Level 1: the bivariate generating function $\mathrm{GK}_c(z,q)$

This refines Level 0 by tracking the max statistic alongside the size:

$$\mathrm{GK}_c(z,q) = \sum_{\mathfrak{c}\in\mathrm{CPP}(c)} z^{\max(\mathfrak{c})}\, q^{|\mathfrak{c}|}.$$

This is the level at which the conjecture is formulated. The variable z grades by the largest entry, decomposing $\mathrm{GK}_c(1,q)$ into the pieces $g_{n,c}(q) = [z^n]\,G_c(z,q)$ from which $Q_{n,c}(q)$ is constructed. Setting $z=1$ recovers Level 0.

7.3 Level 2: the multivariate refinement

A cylindric partition $\mathfrak{c} = (\mu^0,\mu^1,\ldots,\mu^{T-1})$ consists of T individual partitions. The *multivariate generating function* (PhD thesis [13], Chapter 3) tracks each slice separately with its own variable:

$$\sum_{\mathfrak{c}\in\mathrm{CPP}(\pi)} z_0^{|\mu^0|}\, z_1^{|\mu^1|}\cdots z_{T-1}^{|\mu^{T-1}|}. \tag{19}$$

The *refined Borodin identity* states that this equals the product (17) with the substitutions:

$$\begin{aligned} z^{nT} &\mapsto z_0^n z_1^n\cdots z_{T-1}^n, \\ z^{j-i+nT} &\mapsto z_0^n\cdots z_i^n\, z_{i+1}^{n+1}\cdots z_j^{n+1}\, z_{j+1}^n\cdots z_{T-1}^n && (i<j), \\ z^{j-i+(n+1)T} &\mapsto z_0^{n+1}\cdots z_j^{n+1}\, z_{j+1}^n\cdots z_i^n\, z_{i+1}^{n+1}\cdots z_{T-1}^{n+1} && (i>j). \end{aligned} \tag{20}$$

Setting $z_0 = z_1 = \cdots = z_{T-1} = 1$ recovers Level 0. Level 1 ($\mathrm{GK}_c(z,q)$) is obtained from Level 2 by the non-linear operation of replacing $z_0^{|\mu^0|}\cdots z_{T-1}^{|\mu^{T-1}|}$ by $z^{\max_k \mu_1^k} q^{|\mathfrak{c}|}$.

7.4 Level 3: the operator trace (most general Schur form)

The proof in the PhD thesis [13] actually establishes an identity in *free parameters* $u_0,u_1,\ldots,u_T$ and w, from which the multivariate version is a specialisation.

The left-hand side is expressed as a *trace* of operators acting on the ring of symmetric functions:

$$\sum_{\mathfrak{c}\in\mathrm{CPP}(\pi)}\prod_{k=0}^{T-1} z_k^{|\mu^k|} = \sum_{\mu}\langle s_\mu \mid G^{\pi_0}(u_0)\,G^{\pi_1}(u_1)\cdots G^{\pi_{T-1}}(u_{T-1})\,D_w \mid s_\mu\rangle, \tag{21}$$

where $G^1(u) = \Omega[Xu]$ (strip addition), $G^0(u) = \Omega^*[Xu]$ (strip removal), D_w is the degree operator $D_w\, s_\lambda = w^{|\lambda|}\, s_\lambda$, and

$$w = z_0 z_1\cdots z_{T-1}, \qquad u_k = \begin{cases} z_0 z_1\cdots z_{k-1} & \text{if } \pi_k = 1, \\ z_0^{-1} z_1^{-1}\cdots z_{k-1}^{-1} & \text{if } \pi_k = 0. \end{cases} \tag{22}$$

The proof proceeds in three steps:

1. **Commute** all Ω^* operators past all Ω operators using $\Omega^*[Xu]\,\Omega[Xv] = \frac{1}{1-uv}\,\Omega[Xv]\,\Omega^*[Xu]$, producing scalar factors $1/(1-u_iu_j)$ for each pair with $\pi_i > \pi_j$ and $i < j$.

2. **Shift** using $D_w\,\Omega[Xu] = \Omega[Xuw]\,D_w$ and iterate, defining

$$M_\pi(m) = \sum_\mu \langle s_\mu \mid \prod_{k:\,\pi_k=0} \Omega[Xu_kw^m] \prod_{k:\,\pi_k=1} \Omega^*[Xu_k]\,D_w \mid s_\mu\rangle, \tag{23}$$

   which satisfies the recurrence

$$M_\pi(m) = \prod_{\substack{(i,j)\\ \pi_i\neq\pi_j}} \frac{1}{1-u_iu_jw^{m+1}} \;\cdot\; M_\pi(m+1). \tag{24}$$

3. **Take the limit** $m \to \infty$, where $M_\pi(\infty) = \prod_{n\geq 1} 1/(1-w^n)$.

The result is an identity in the free parameters $u_0, \ldots, u_{T-1}, w$:

$$\operatorname{tr}\big(G^{\pi_0}(u_0)\cdots G^{\pi_{T-1}}(u_{T-1})\,D_w\big) = \prod_{m\geq 1}\frac{1}{1-w^m} \prod_{\substack{i<j\\ \pi_i>\pi_j}} \frac{1}{1-u_iu_jw^{m-1}} \prod_{\substack{i>j\\ \pi_i>\pi_j}} \frac{1}{1-u_iu_jw^m}. \tag{25}$$

The multivariate identity (Level 2) is recovered by the specialisation (21), and the univariate identity (Level 0) by the further specialisation $z_k \mapsto z$ for all k.

7.5 Level 4: the (r,t)-Macdonald extension

The richest form replaces the Schur inner product by the Macdonald inner product and each cylindric partition by a weighted object. For $\mathfrak{c} = (\mu^0, \mu^1, \ldots, \mu^T)$, define the *Macdonald weight*

$$W_\mathfrak{c}(r,t) := \prod_{\substack{k=1\\ \pi_k=1}}^{T} \varphi_{\mu^k/\mu^{k-1}}(r,t) \prod_{\substack{k=1\\ \pi_k=0}}^{T} \psi_{\mu^{k-1}/\mu^k}(r,t), \tag{26}$$

where $\varphi_{\lambda/\mu}(r,t)$ and $\psi_{\lambda/\mu}(r,t)$ are the Macdonald Pieri coefficients:

$$\varphi_{\lambda/\mu}(r,t) = \prod_{s\in C_{\lambda/\mu}} \frac{1-r^{a_\lambda(s)}\,t^{\ell_\lambda(s)+1}}{1-r^{a_\lambda(s)+1}\,t^{\ell_\lambda(s)}}, \tag{27}$$

$$\psi_{\lambda/\mu}(r,t) = \prod_{s\notin C_{\lambda/\mu}} \frac{1-r^{a_\lambda(s)+1}\,t^{\ell_\lambda(s)}}{1-r^{a_\lambda(s)}\,t^{\ell_\lambda(s)+1}}, \tag{28}$$

with $a_\lambda(s)$ and $\ell_\lambda(s)$ the arm and leg lengths of the box s in λ, and $C_{\lambda/\mu}$ the set of columns where the skew shape λ/μ has its boxes.

The left-hand side of the (r,t)-identity is expressed as an operator trace over Macdonald polynomials (PhD thesis [13], Chapter 4):

$$\sum_{\mathfrak{c}\in\mathrm{CPP}(\pi)} W_\mathfrak{c}(r,t) \prod_{k=0}^{T-1} z_k^{|\mu^k|} = \sum_\mu \langle Q_\mu \mid H^{\pi_0}(u_0)\,H^{\pi_1}(u_1)\cdots H^{\pi_{T-1}}(u_{T-1})\,D_w \mid P_\mu\rangle_{r,t}, \tag{29}$$

where $H^1(u) = \Omega_{r,t}[Xu]$, $H^0(u) = \Omega^*_{r,t}[Xu]$, and $\langle\,\cdot\,,\,\cdot\,\rangle_{r,t}$ is the Macdonald inner product with respect to which $\{P_\mu\}$ and $\{Q_\mu\}$ are dual bases. The proof is identical in structure to the Schur case, using the (r,t)-commutation relation

$$\Omega^*_{r,t}[Xu]\,\Omega_{r,t}[Xv] = \frac{(tuv;r)_\infty}{(uv;r)_\infty}\,\Omega_{r,t}[Xv]\,\Omega^*_{r,t}[Xu]$$

in place of $\Omega^*[Xu]\,\Omega[Xv] = \frac{1}{1-uv}\,\Omega[Xv]\,\Omega^*[Xu]$.

The (r,t)-recurrence $N_\pi(m) \to N_\pi(m{+}1)$ produces the factors $(tu_iu_jw^{m+1};r)_\infty/(u_iu_jw^{m+1};r)_\infty$, and the same limit $N_\pi(\infty) = \prod_{n\geq 1} 1/(1-w^n)$ gives:

$$\mathrm{tr}\big(H^{\pi_0}(u_0)\cdots H^{\pi_{T-1}}(u_{T-1})\,D_w\big)_{r,t} = \prod_{m\geq 1}\frac{1}{1-w^m}\prod_{\substack{i<j\\ \pi_i>\pi_j}}\frac{(tu_iu_jw^{m-1};r)_\infty}{(u_iu_jw^{m-1};r)_\infty}\prod_{\substack{i>j\\ \pi_i>\pi_j}}\frac{(tu_iu_jw^{m};r)_\infty}{(u_iu_jw^{m};r)_\infty}. \tag{30}$$

The weight function has a combinatorial description in terms of *peak* and *valley* cubes (PhD thesis, Theorem 4.2):

$$W_{\mathfrak{c}}(r,t) = \Omega\big[(r-t)\,\mathcal{D}_{\mathfrak{c}}(r,t)\big]\,, \tag{31}$$

where $\Omega[\cdot]$ is the plethystic exponential and

$$\mathcal{D}_{\mathfrak{c}}(r,t) = \sum_{s\in\mathrm{peak}(\mathfrak{c})} r^{a_{\mathfrak{c}}(s)}\,t^{\ell_{\mathfrak{c}}(s)} - \sum_{s\in\mathrm{valley}(\mathfrak{c})} r^{a_{\mathfrak{c}}(s)}\,t^{\ell_{\mathfrak{c}}(s)}.$$

7.6 Summary: the nesting

The five levels form a chain of specialisations:

Level 4 — (r,t)-operator trace in free parameters u_k, w *(most general)*:

$$\mathrm{tr}\big(H^{\pi_0}(u_0)\cdots H^{\pi_{T-1}}(u_{T-1})\,D_w\big)_{r,t} = \prod_{m\geq 1}\frac{1}{1-w^m}\prod_{\pi_i>\pi_j}\frac{(tu_iu_jw^\bullet;r)_\infty}{(u_iu_jw^\bullet;r)_\infty}$$

$\Big\downarrow$ *set $r=t$ (all Macdonald weights $\to 1$)*

Level 3 — Schur operator trace in free parameters u_k, w:

$$\mathrm{tr}\big(G^{\pi_0}(u_0)\cdots G^{\pi_{T-1}}(u_{T-1})\,D_w\big) = \prod_{m\geq 1}\frac{1}{1-w^m}\prod_{\pi_i>\pi_j}\frac{1}{1-u_iu_jw^\bullet}$$

$\Big\downarrow$ *specialise u_k, w via* (21)

Level 2 — Multivariate: $\displaystyle\sum_{\mathfrak{c}} z_0^{|\mu^0|}z_1^{|\mu^1|}\cdots z_{T-1}^{|\mu^{T-1}|}$

$\Big\downarrow$ *set all $z_k=1$* $\qquad\qquad$ $\Big\downarrow$ *max extraction $\to z^{\max_k \mu_1^k}q^{|\mathfrak{c}|}$*

Level 0 — Size: $\displaystyle\sum_{\mathfrak{c}} q^{|\mathfrak{c}|}$ $\qquad\qquad$ **Level 1** — Bivariate: $\mathrm{GK}_c(z,q)=\displaystyle\sum_{\mathfrak{c}} z^{\max(\mathfrak{c})}q^{|\mathfrak{c}|}$

$\Big\downarrow$ *set $z=1$*

Level 0 — Size: $\displaystyle\sum_{\mathfrak{c}} q^{|\mathfrak{c}|}$

Level 1 (the bivariate $\mathrm{GK}_c(z,q)$) is the level at which Conjecture 9 is stated. It sits "beside" Level 2 rather than strictly below it, because the max statistic is obtained from the multivariate data by a non-linear operation (taking the maximum of the first parts).

8 The Macdonald extension of the conjecture

8.1 Notation

The Macdonald polynomials $P_\lambda(X;r,t)$ depend on two parameters traditionally called q and t. Since we have already used q as the weight-tracking variable throughout this document (as in $q^{|\mathfrak{c}|}$ and $Q_{n,c}(q)$), we adopt the convention of writing (r,t) for the Macdonald parameters to avoid ambiguity. The specialisation $r=t$ recovers the Schur function setting.

8.2 The (r,t)-extended generating function

Define the *Macdonald-weighted bivariate generating function*:

$$\mathrm{GK}_c^{(r,t)}(z,q) := \sum_{\mathfrak{c}\in\mathrm{CPP}(c)} W_{\mathfrak{c}}(r,t)\cdot z^{\max(\mathfrak{c})}\,q^{|\mathfrak{c}|}, \tag{32}$$

where the Macdonald weight $W_{\mathfrak{c}}(r,t)$ is as defined in (25) (with the substitution $q\mapsto r$):

$$W_{\mathfrak{c}}(r,t) = \prod_{k:\,\pi_k=1}\varphi_{\mu^k/\mu^{k-1}}(r,t)\prod_{k:\,\pi_k=0}\psi_{\mu^{k-1}/\mu^k}(r,t).$$

This specialises to the ordinary generating function when $r = t$:

$$\mathrm{GK}_c^{(r,r)}(z,q) = \mathrm{GK}_c(z,q).$$

8.3 The corrected coefficients

As in the Schur case, define the corrected generating function and its z-coefficients:

$$G_c^{(r,t)}(z,q) := (zq;q)_\infty \cdot \mathrm{GK}_c^{(r,t)}(z,q), \tag{33}$$

$$g_{n,c}^{(r,t)}(q) := [z^n]\, G_c^{(r,t)}(z,q). \tag{34}$$

Note that the E–H factorisation uses $(zq;q)_\infty$, which depends only on the weight variable q and *not* on the Macdonald parameters (r,t). This is because the "universal factor" $H(z) = 1/(zq;q)_\infty$ counts ordinary partitions by size, independently of any Macdonald weighting.

8.4 The polynomial $Q_{n,c}(q;\, r,t)$

Definition 13. For a nonneg integer n, profile $c = (c_0, c_1, c_2)$ with $d = c_0 + c_1 + c_2$, $d \not\equiv 0 \pmod 3$, and Macdonald parameters (r,t), define

$$Q_{n,c}(q;\, r,t) := (q;q)_n \cdot g_{n,c}^{(r,t)}(q) = (q;q)_n \cdot [z^n]\Big((zq;q)_\infty \cdot \mathrm{GK}_c^{(r,t)}(z,q)\Big). \tag{35}$$

By construction, $Q_{n,c}(q;\, r,r) = Q_{n,c}(q)$.

Remark 14 (Structure of the coefficients)**.** Since the Macdonald Pieri coefficients $\varphi_{\lambda/\mu}(r,t)$ and $\psi_{\lambda/\mu}(r,t)$ are rational functions of r and t, the quantity $Q_{n,c}(q;\, r,t)$ is *a priori* a formal power series in q with coefficients in $\mathbb{Q}(r,t)$. For it to be a polynomial in q with "positive" coefficients, we would need:

(a) $Q_{n,c}(q;\, r,t)$ is a polynomial in q (finitely many nonzero q-powers), and

(b) each q-coefficient is a nonneg expression in r and t.

The precise meaning of "nonneg in r,t" requires care: the Pieri coefficients are ratios of terms of the form $(1 - r^a t^b)$, so the natural positivity domain is $0 < r, t < 1$.

8.5 The conjecture

Conjecture 15 (Macdonald extension)**.** *Let $c = (c_0, c_1, c_2)$ with $d := c_0 + c_1 + c_2$ and $d \not\equiv 0 \pmod 3$. Then:*

(i) $Q_{n,c}(q;\, r,t)$ is a polynomial in q whose coefficients are rational functions of r,t that are ***nonneg*** *for $0 < r, t < 1$.*

(ii) Specialising $r = t$ recovers Conjecture 9:

$$Q_{n,c}(q;\, r,r) = Q_{n,c}(q).$$

(iii) At $q = 1$:

$$Q_{n,c}(1;\, r,t) = \left(\frac{(d+1)(d+2)}{6} - 1\right)^n,$$

independent of (r,t).

Remark 16 (Why (iii) should be independent of (r,t))**.** The evaluation at $q=1$ counts cylindric partitions at the "combinatorial" level (ignoring the q-grading), and the Macdonald weight satisfies $W_{\mathfrak{c}}(r,t)\big|_{q=1}=1$ for any cylindric partition $\mathfrak{c}$, since the Pieri coefficients collapse: the products of $(1-r^at^b)$ factors in numerator and denominator cancel pairwise when the q-weight is forgotten. More precisely, summing the weight variable $q^{|\mathfrak{c}|}$ at $q=1$ eliminates the distinction between different q-refinements, so the $q=1$ evaluation depends only on the combinatorial count.

Remark 17 (The (r,t)-Borodin product at $z=1$)**.** Setting $z=1$ in $\mathrm{GK}_c^{(r,t)}(z,q)$ gives a (r,t)-deformed character. Via the (r,t)-Borodin identity (29), this is:

$$\mathrm{GK}_c^{(r,t)}(1,q)=\prod_{n\geq 1}\frac{1}{1-q^n}\prod_{\substack{i<j\\ \pi_i>\pi_j}}\frac{(tq^{j-i+(n-1)T};r)_\infty}{(q^{j-i+(n-1)T};r)_\infty}\prod_{\substack{i>j\\ \pi_i>\pi_j}}\frac{(tq^{j-i+nT};r)_\infty}{(q^{j-i+nT};r)_\infty},$$

where $(a;r)_\infty=\prod_{k\geq 0}(1-ar^k)$. At $r=t$ the (r,t)-factors reduce to 1 and we recover the ordinary Borodin product.

Remark 18 (Why the conjecture is natural)**.** Each individual Macdonald weight $W_{\mathfrak{c}}(r,t)$ is a product of positive rational functions (for $0<r,t<1$), so each cylindric partition contributes a positive term to $\mathrm{GK}_c^{(r,t)}(z,q)$. The alternating-sign cancellation in the E–H factorisation (multiplication by $(zq;q)_\infty$) is the same Euler expansion $\sum(-1)^jq^{j(j+1)/2}z^j/(q)_j$ as in the Schur case. If the cancellation resolves into positivity for $r=t$ (the original conjecture), it is natural to expect it does so for general (r,t) as well, since the Macdonald weights only *redistribute* the mass among cylindric partitions without changing the total count.

9 Multivariate extensions

The conjecture as stated in Section 4 uses the bivariate generating function $\mathrm{GK}_c(z,q)$, which tracks the single statistic $\max(\mathfrak{c})$ via the variable z. At Level 2 in the hierarchy (Section 7), we have the multivariate generating function with separate variables $z_0,z_1,\ldots,z_{T-1}$ tracking $|\mu^0|,|\mu^1|,\ldots,|\mu^{T-1}|$ individually. This suggests natural multivariate lifts of the conjecture.

9.1 Multivariate E–H factorisation

For rank 3 with profile $c=(c_0,c_1,c_2)$, the cylindric partition $\mathfrak{c}=(\mu^0,\mu^1,\mu^2)$ consists of three partitions. Define the *multivariate generating function*

$$\mathrm{GK}_c(\mathbf{z},q):=\sum_{\mathfrak{c}\in\mathrm{CPP}(c)}z_0^{|\mu^0|}\,z_1^{|\mu^1|}\,z_2^{|\mu^2|}\,q^{|\mathfrak{c}|},$$

where $q^{|\mathfrak{c}|}=q^{|\mu^0|+|\mu^1|+|\mu^2|}$. (Here $\mathbf{z}=(z_0,z_1,z_2)$ and the q-variable is redundant: $q^{|\mathfrak{c}|}=(z_0z_1z_2)^?\cdot q^?$ is already encoded in the z-variables. More precisely, one can set $q=1$ and work purely with z_0,z_1,z_2, or set $z_0=z_1=z_2=q$ to recover $\mathrm{GK}_c(q)=\mathrm{GK}_c(1,q)$.)

For the multivariate E–H factorisation, define the corrected generating function

$$G_c(\mathbf{z}):=(z_0z_1z_2;z_0z_1z_2)_\infty\cdot\mathrm{GK}_c(\mathbf{z},1),\tag{36}$$

and its multi-index coefficients

$$g_{\mathbf{n},c}:=[z_0^{n_0}z_1^{n_1}z_2^{n_2}]\,G_c(\mathbf{z}),\qquad\mathbf{n}=(n_0,n_1,n_2).\tag{37}$$

9.2 The multivariate polynomial $Q_{\mathbf{n},c}$

Definition 19 (Multivariate Q)**.** For a multi-index $\mathbf{n} = (n_0, n_1, n_2) \in \mathbb{N}^3$, profile $c = (c_0, c_1, c_2)$ with $d = c_0 + c_1 + c_2$, $d \not\equiv 0 \pmod 3$, define

$$Q_{\mathbf{n},c} := \prod_{k=0}^{2} (q;q)_{n_k} \cdot g_{\mathbf{n},c}(q), \tag{38}$$

where $q = z_0 z_1 z_2$ and $g_{\mathbf{n},c}(q)$ is the appropriately normalised multi-index coefficient.

Remark 20. Setting $n_0 = n_1 = n_2 = n$ and extracting the coefficient of $z_0^n z_1^n z_2^n$ from the multivariate generating function is *not* the same as the bivariate $[z^n]$ extraction (which uses the max statistic). The relationship between the multivariate and bivariate conjectures is therefore non-trivial: the bivariate conjecture concerns a different projection of the same underlying data.

Conjecture 21 (Multivariate positivity)**.** *Let $c = (c_0, c_1, c_2)$ with $d := c_0 + c_1 + c_2$ and $d \not\equiv 0 \pmod 3$. Then for all $\mathbf{n} = (n_0, n_1, n_2) \in \mathbb{N}^3$:*

*(i) $Q_{\mathbf{n},c}$ is a polynomial in q with **nonneg coefficients**.*

(ii) Specialising to the diagonal $n_0 = n_1 = n_2 = n$ and projecting via the max statistic recovers Conjecture 9.

9.3 The (r,t)-Macdonald multivariate conjecture

Combining the Macdonald extension (Section 8) with the multivariate framework, define the *Macdonald-weighted multivariate generating function*:

$$\mathrm{GK}_c^{(r,t)}(\mathbf{z}) := \sum_{\mathfrak{c} \in \mathrm{CPP}(c)} W_{\mathfrak{c}}(r,t) \cdot z_0^{|\mu^0|}\, z_1^{|\mu^1|}\, z_2^{|\mu^2|},$$

and the corrected multi-index coefficients

$$g_{\mathbf{n},c}^{(r,t)} := [z_0^{n_0} z_1^{n_1} z_2^{n_2}]\Big((z_0 z_1 z_2; z_0 z_1 z_2)_\infty \cdot \mathrm{GK}_c^{(r,t)}(\mathbf{z}) \Big). \tag{39}$$

Definition 22 (Macdonald multivariate Q)**.** For a multi-index $\mathbf{n} = (n_0, n_1, n_2) \in \mathbb{N}^3$, define

$$Q_{\mathbf{n},c}(q;\, r,t) := \prod_{k=0}^{2} (q;q)_{n_k} \cdot g_{\mathbf{n},c}^{(r,t)}(q), \tag{40}$$

where $q = z_0 z_1 z_2$.

Conjecture 23 (Macdonald multivariate positivity)**.** *Let $c = (c_0, c_1, c_2)$ with $d := c_0 + c_1 + c_2$ and $d \not\equiv 0 \pmod 3$. Then for all $\mathbf{n} = (n_0, n_1, n_2) \in \mathbb{N}^3$:*

*(i) $Q_{\mathbf{n},c}(q;\, r,t)$ is a polynomial in q whose coefficients are rational functions of r,t that are **nonneg** for $0 < r, t < 1$.*

(ii) Specialising $r = t$ recovers Conjecture 21:

$$Q_{\mathbf{n},c}(q;\, r,r) = Q_{\mathbf{n},c}(q).$$

(iii) Restricting to the bivariate setting (single index n via the max statistic) recovers Conjecture 15.

Remark 24 (Chain of implications)**.** The three conjectures form a chain of increasing generality:

$$\text{Conjecture 23} \underset{r=t}{\Longrightarrow} \text{Conjecture 21} \underset{\text{max proj.}}{\Longrightarrow} \text{Conjecture 9}$$

and the bivariate Macdonald version fits as:

$$\text{Conjecture 23} \underset{\text{max proj.}}{\Longrightarrow} \text{Conjecture 15} \underset{r=t}{\Longrightarrow} \text{Conjecture 9.}$$

10 Cylindric Kostka polynomials

The polynomials $Q_{n,c}(q)$ bear a striking structural resemblance to the classical Kostka–Foulkes polynomials $K_{\lambda,\mu}(t)$. We argue that they should be regarded as *cylindric* analogues of Kostka polynomials, and that this perspective clarifies both the conjecture and the path toward its proof.

10.1 Recollection: Kostka polynomials

The *Kostka–Foulkes polynomials* $K_{\lambda,\mu}(t)$ [15] arise as the transition coefficients between the Schur and Hall–Littlewood bases of the ring of symmetric functions:

$$s_\lambda(X) = \sum_\mu K_{\lambda,\mu}(t)\, P_\mu(X;t), \tag{41}$$

where $P_\mu(X;t)$ is the Hall–Littlewood polynomial. Equivalently, inverting the relation, $P_\mu(X;t) = \sum_\lambda \widetilde{K}_{\lambda,\mu}(t)\, s_\lambda(X)$ expresses the Hall–Littlewood polynomial in the Schur basis.

Their key properties are:

(K1) **Polynomiality:** $K_{\lambda,\mu}(t) \in \mathbb{Z}[t]$.

(K2) **Positivity:** $K_{\lambda,\mu}(t)$ has nonneg integer coefficients (Lascoux–Schützenberger [14]).

(K3) **Combinatorial interpretation:** $K_{\lambda,\mu}(t) = \sum_T t^{\mathrm{charge}(T)}$, summed over semistandard Young tableaux T of shape λ and weight μ, where charge is Lascoux and Schützenberger's charge statistic.

(K4) **Evaluation:** $K_{\lambda,\mu}(1) = K_{\lambda,\mu}$, the Kostka number counting semistandard Young tableaux of shape λ and weight μ.

(K5) **Macdonald lift:** There exist two-parameter polynomials $K_{\lambda,\mu}(r,t)$ (the *Macdonald–Kostka polynomials* or (r,t)-Kostka polynomials) with $K_{\lambda,\mu}(0,t) = K_{\lambda,\mu}(t)$.

(K6) **Representation theory:** The Kostka polynomials encode multiplicities in the decomposition of representations of GL_n and appear as Kazhdan–Lusztig polynomials for the affine Weyl group of type $\mathrm{A}_1^{(1)}$.

10.2 The cylindric analogue

The polynomials $Q_{n,c}(q)$ share each of these properties (proved or conjectured):

(C1) **Polynomiality:** $Q_{n,c}(q) \in \mathbb{Z}[q]$ (proved by Welsh [12]).

(C2) **Positivity:** $Q_{n,c}(q)$ has nonneg integer coefficients (Conjecture 9, open).

(C3) **Combinatorial interpretation:** Unknown. Finding a charge-like statistic on cylindric partitions (or cylindric tableaux) is the central open problem.

(C4) **Evaluation:** $Q_{n,c}(1) = \left(\frac{(d+1)(d+2)}{6} - 1\right)^n$ (proved by Welsh [12]).

(C5) **Macdonald lift:** The (r,t)-extended polynomial $Q_{n,c}(q;\, r,t)$ of Definition 13, with $Q_{n,c}(q;\, r,r) = Q_{n,c}(q)$.

(C6) **Representation theory:** The $Q_{n,c}(q)$ appear as numerators in $\mathrm{A}_2^{(1)}$ Andrews–Gordon characters via equation (16).

Definition 25 (Cylindric Kostka polynomial)**.** We call $Q_{n,c}(q)$ the *cylindric Kostka polynomial* of index n and profile c. More generally, $Q_{\mathbf{n},c}(q;\, r,t)$ (Definition 22) is the (r,t)-*cylindric Kostka polynomial* of multi-index $\mathbf{n}$ and profile c.

10.3 Schur expansion of the Borodin product

The multivariate Borodin product (Level 2, Section 7) is a formal power series in z_0, z_1, z_2:

$$\mathrm{GK}_c(\mathbf{z}) = \sum_{\mathfrak{c} \in \mathrm{CPP}(c)} z_0^{|\mu^0|}\, z_1^{|\mu^1|}\, z_2^{|\mu^2|} = \prod (\text{Borodin factors}).$$

This product can be expanded in terms of Schur functions $s_\lambda(z_0, z_1, z_2)$ (for partitions λ with at most 3 parts):

$$\mathrm{GK}_c(\mathbf{z}) = \sum_{\lambda} a_{\lambda,c}\, s_\lambda(z_0, z_1, z_2), \tag{42}$$

where the coefficients $a_{\lambda,c}$ are formal power series.

Remark 26. The multivariate generating function $\mathrm{GK}_c(\mathbf{z})$ is *not* symmetric in (z_0, z_1, z_2) in general (the profile c breaks the symmetry unless $c_0 = c_1 = c_2$). The Schur functions $\{s_\lambda\}_{\ell(\lambda)\le 3}$ form a basis for the *symmetric* polynomials in three variables, so the expansion (41) captures the symmetric part. The full expansion requires additional terms from the non-symmetric sector—for instance, Demazure characters (key polynomials) or the monomial basis. For a cyclic profile ($c_0 = c_1 = c_2$), the generating function is symmetric and the Schur expansion suffices.

The coefficients $a_{\lambda,c}$ in (41) are not known in closed form. One might hope that non-negativity of these Schur expansion coefficients would imply the positivity conjecture for $Q_{n,c}(q)$. However, the relationship between the two is **not straightforward**, for two reasons:

1. **Max extraction.** The bivariate generating function $\mathrm{GK}_c(z,q)$ is obtained from the multivariate one by the non-linear operation $z_0^{|\mu^0|} z_1^{|\mu^1|} z_2^{|\mu^2|} \mapsto z^{\max_k \mu_1^k}\, q^{|\mathfrak{c}|}$. This is *not* a linear map on formal power series, so Schur-positivity of $\mathrm{GK}_c(\mathbf{z})$ does not directly transfer to the bivariate setting.

2. *E–H* **factorisation.** Even after passing to the bivariate $\mathrm{GK}_c(z,q)$, the construction of $Q_{n,c}(q)$ involves multiplication by $(zq;q)_\infty = \sum_{j\geq 0}(-1)^j q^{j(j+1)/2} z^j/(q)_j$, which introduces **alternating signs**. The conjecture asserts that these alternating-sign cancellations resolve perfectly into non-negativity after clearing denominators—but this cancellation is the essential difficulty, and it is not implied by any positivity of the Schur expansion coefficients $a_{\lambda,c}$.

In other words, the Schur expansion coefficients of the Borodin product and the cylindric Kostka polynomials $Q_{n,c}(q)$ are **distinct** objects. The name "cylindric Kostka polynomial" for $Q_{n,c}(q)$ is justified not because $Q_{n,c}(q)$ *is* a Schur expansion coefficient, but because it plays the same *structural role*: it is the polynomial numerator that appears when an affine Lie algebra character is decomposed as $\sum Q_{n,c}(q)/(q)_n$.

Remark 27 (The $\mathrm{A}_1 \to \mathrm{A}_2$ step)**.** In the A_1 setting (rank 2, one profile entry), the Andrews–Gordon identity has the form $\sum_n q^{f(n)}/(q)_n$ = product, where $f(n)$ is a quadratic form. The "Kostka polynomial" is the monomial $q^{f(n)}$—trivially positive.

The A_2 step replaces this monomial by a genuine polynomial $Q_{n,c}(q)$ with (conjecturally) many terms, all positive. This is the essential new phenomenon: at rank 3, the cylindric Kostka polynomials are *not* monomials (except for $d = 2$, which reduces to the A_1 case).

10.4 The missing charge statistic

The single most important open problem is finding the combinatorial counterpart of property (C3): a *charge-like statistic* on some set of cylindric objects.

Problem 28. Find a finite set $\mathcal{T}_{n,c}$ with $|\mathcal{T}_{n,c}| = Q_{n,c}(1) = \left(\frac{(d+1)(d+2)}{6} - 1\right)^n$ and a statistic $\mathrm{ch}\colon \mathcal{T}_{n,c} \to \mathbb{N}$ such that

$$Q_{n,c}(q) = \sum_{T \in \mathcal{T}_{n,c}} q^{\mathrm{ch}(T)}. \tag{43}$$

The natural candidates for $\mathcal{T}_{n,c}$ include:

- **Cylindric tableaux** — semistandard fillings of a cylindric shape, generalising the SSYT that index ordinary Kostka polynomials.
- **Paths in the alcove geometry** — the $\mathrm{A}_2^{(1)}$ alcove walk model, which provides the representation-theoretic setting for the conjecture.
- **Cylindric partitions with bounded max** — the set $\mathrm{CPP}(c)_{\max=n}$ has the right generating function (before clearing denominators), and the cyclic descent injection studied in [13] provides a partial weight-raising bijection on this set.

Solving Problem 28 would simultaneously prove Conjecture 9 and provide a manifestly positive multisum formula for every A_2 Andrews–Gordon identity.

11 Computational evidence

The conjecture has been verified computationally for:

d	$Q_{n,c}(1)$	Profiles tested	Max n verified
1	$1^n = 1$	$(1,0,0)$	all n (trivial)
2	2^n	$(2,0,0)$, $(1,1,0)$	all n (Q is monomial)
4	4^n	$(2,2,0)$, $(2,1,1)$	$n \leq 3$
5	6^n	$(3,1,1)$, $(3,2,0)$, $(2,2,1)$	$n \leq 3$
7	11^n	$(3,2,2)$, $(2,3,2)$, $(4,2,1)$	$n \leq 2$
8	14^n	$(3,3,2)$, $(4,3,1)$, $(4,2,2)$	$n \leq 1$

In every case, all coefficients of $Q_{n,c}(q)$ are nonneg. Moreover, the corrected coefficients $h_{m,c}(q) = (q;q)_m \cdot g_{m,c}(q)$ are themselves nonneg polynomials, which is a strictly stronger observation.

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